

Nacham Sathvik

Embedded Systems Engineer

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PROFESSIONAL SUMMARY

Embedded Systems Engineer with 2+ years of experience in firmware development, low-power IoT design, and wireless communication systems. Skilled in Embedded C, RTOS, and multi-protocol interfacing (UART, SPI, I2C, RS485, Modbus). Delivered end-to-end embedded projects — from PCB-level hardware integration to cloud data transmission — at Amptronics Systems Pvt. Ltd., Hyderabad.

WORK EXPERIENCE

Embedded Systems Engineer — Amptronics Systems Pvt. Ltd.

Dec 2023 – Present | Hyderabad

- Developed and maintained firmware for ARM Cortex-M3, ESP32, STM32, and STM8 microcontrollers using Embedded C with RTOS task scheduling.
- Interfaced multiple sensors and peripherals over UART, SPI, I2C, and RS485 on custom-designed PCBs, ensuring reliable data acquisition.
- Reduced system current consumption by ~30% through RTOS sleep modes and peripheral gating, extending battery life by up to 2x on battery-powered sensor nodes.
- Built RS485 IoT gateway transmitting real-time data to cloud via MQTT/TCP-IP, enabling remote monitoring for clients.
- Configured interrupt handlers (ISRs) for GPIO, UART, and timer peripherals, and managed RTOS tasks, queues, and semaphores to ensure reliable multi-task execution.
- Implemented watchdog timer for system fault recovery and performed firmware debugging using JTAG/SWD, identifying and resolving critical issues on production hardware.
- Implemented Modbus RTU protocol on RS485 bus to poll and control industrial slave devices with reliable data exchange.
- Developed cloud connectivity firmware on ESP32 using MQTT protocol with auto-reconnect handling for reliable data publishing in production IoT systems.

KEY PROJECTS

RF-Based Wireless Data Transmission System

- Developed firmware for reliable RF-based point-to-point data transfer, achieving low packet error rate in industrial environments.

CT-Based Self-Powered Wireless Sensor

- Designed firmware for a battery-free sensor node powered by current transformer energy harvesting, eliminating maintenance downtime.

RS485 Industrial IoT Gateway — Cloud Data Transmission

- Built RS485 master-slave gateway supporting multiple devices with MQTT/TCP-IP cloud uplink, enabling real-time remote monitoring for industrial clients.

RS485 Data Logging System

- Implemented structured data logging over RS485 network for industrial devices which support RS485, supporting reliable data retrieval and audit trails.

TECHNICAL SKILLS

Microcontrollers: ARM Cortex-M3, ESP32, STM32, STM8, 8051, ATmega

Programming Languages: Embedded C, C, Python, Assembly

Protocols & Interfaces: UART, SPI, I2C, RS485, Modbus, MQTT, TCP/IP

Wireless: LoRa, GSM

Dev Tools: STM32 CubeIDE, Keil uVision, Arduino IDE, ESP-IDF, Code Composer Studio

Debug & Test Tools: Logic Analyzer, Oscilloscope, Protocol Decoder, Smart RF Studio

EDUCATION

B.Tech — Electronics and Communication Engineering	2023 7.0 CGPA
<i>Siddhartha Institute of Engineering and Technology, JNTU-H</i>	
Intermediate (MPC)	2019 86.3%
<i>Narayana Junior College</i>	
SSC	2017 8.7 CGPA
<i>Krishnaveni Talent School</i>	

CERTIFICATIONS & ACHIEVEMENTS

- International Conference on Modern Computing in Circuits and Systems 2023 — Certification.